

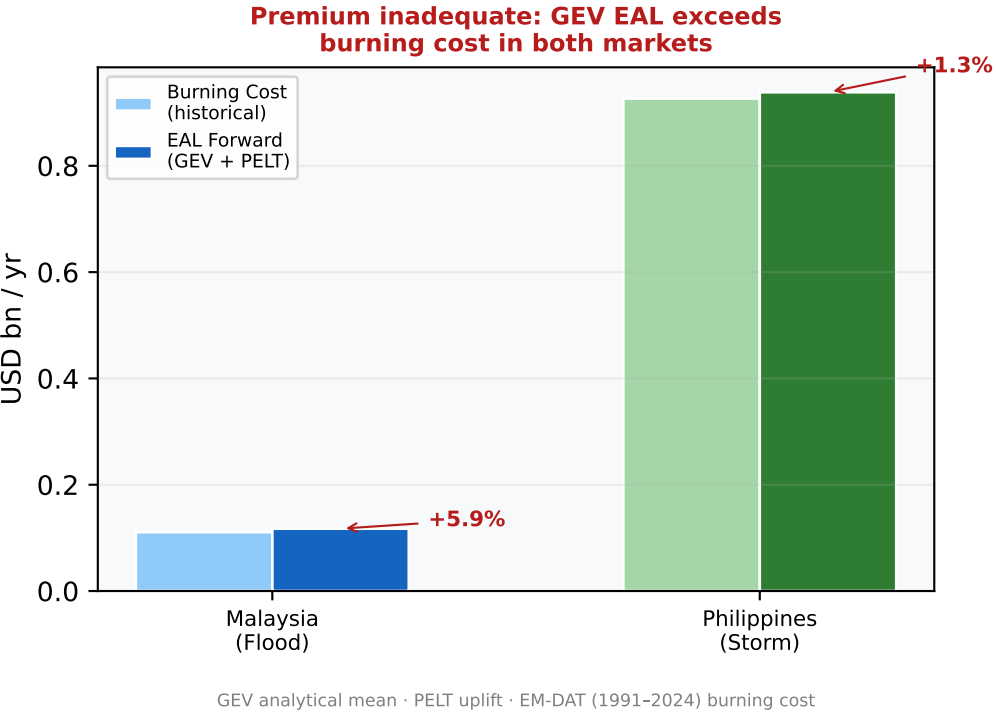
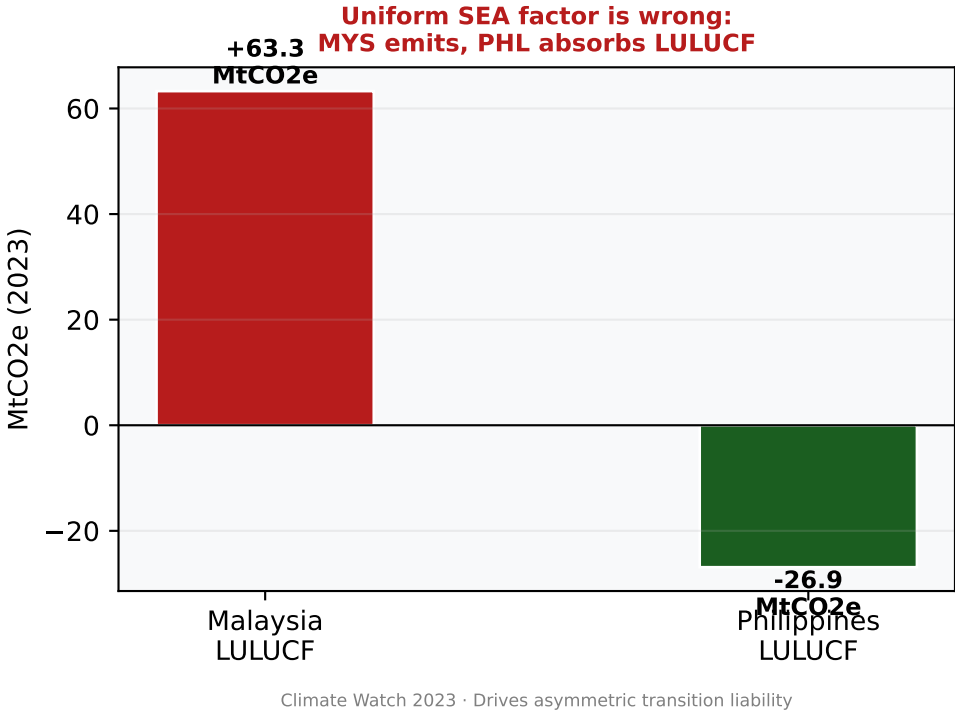
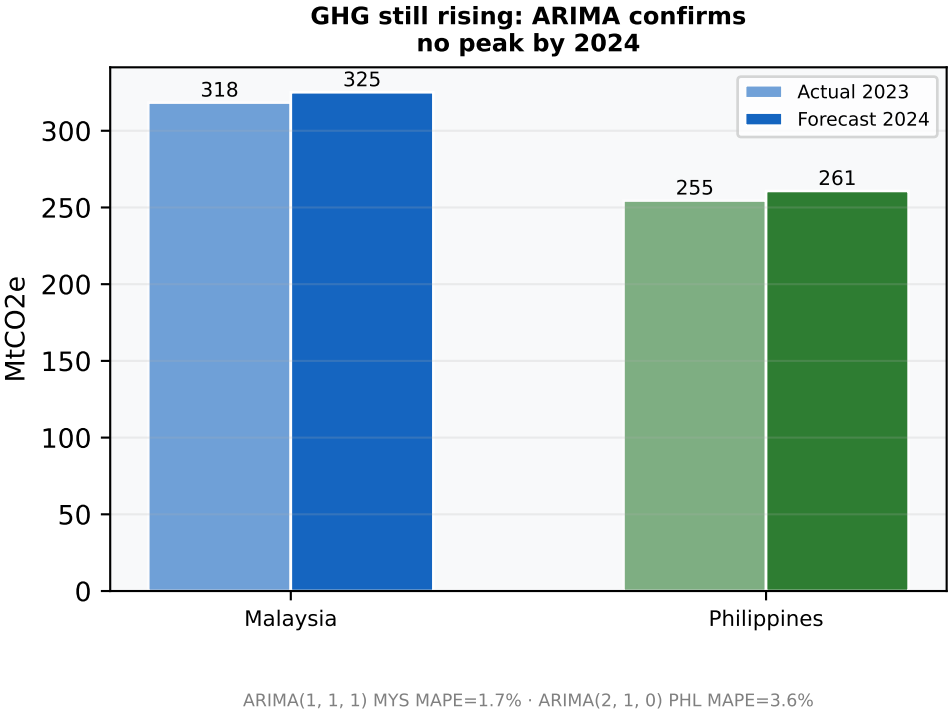
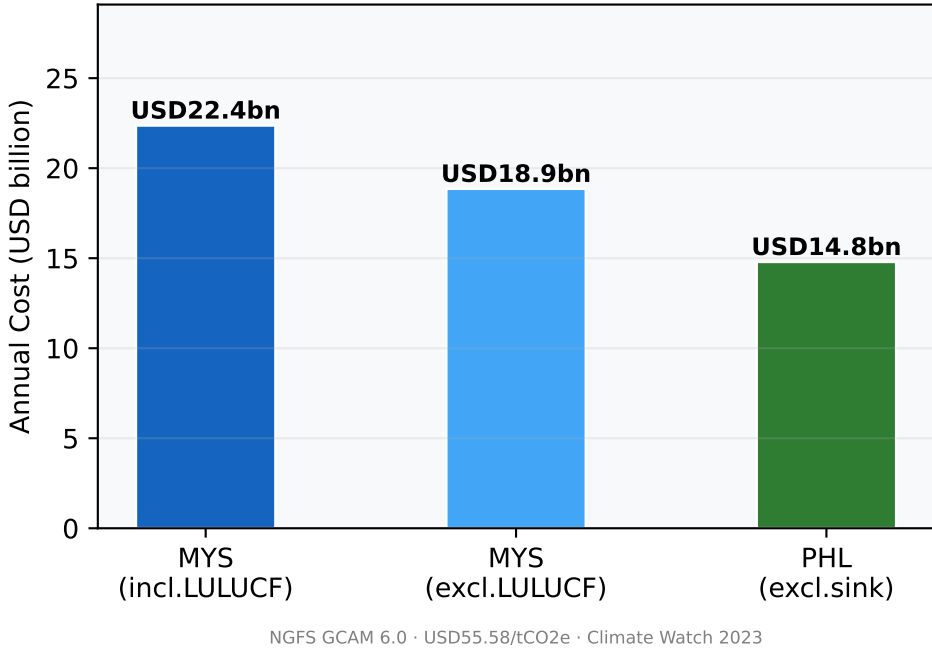
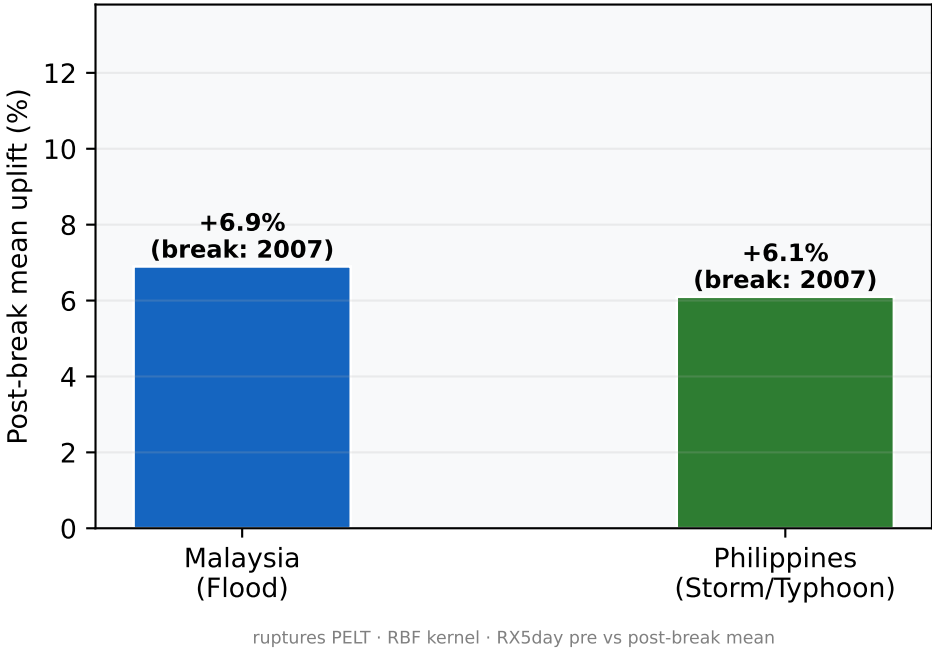
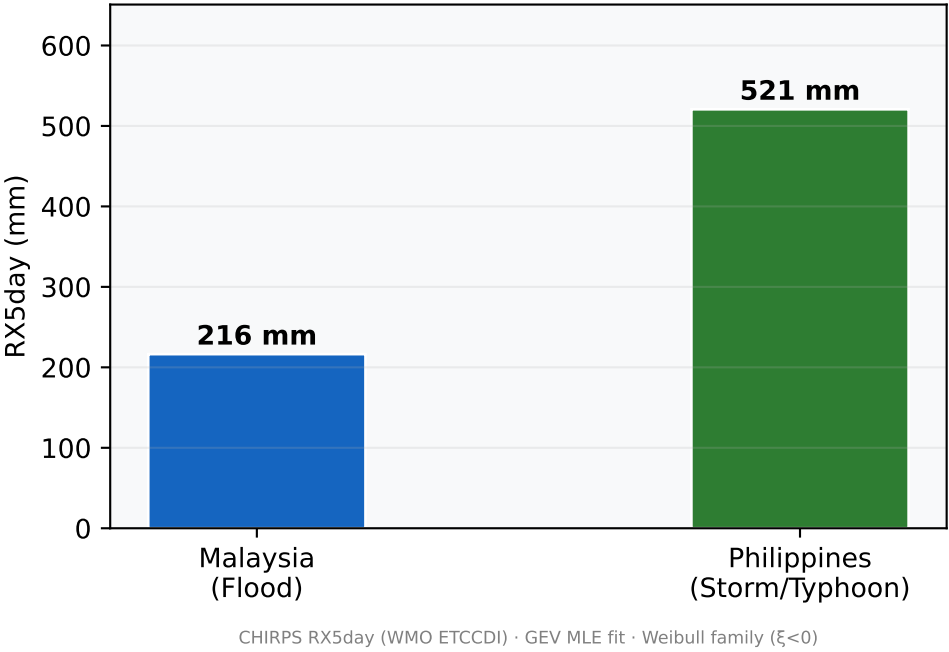
Hannover Re SEA Treaty Book — Physical & Transition Risk · Three-Action Framework

R5 Executive Summary | Malaysia & Philippines | Pre-2007 Hazard Transition Risk Quantification

Physical gap: 100-yr flood is structurally underpriced

Regime shift: post-2007 hazard baseline +6-7% unreserved

Transition gap: MYS faces USD 22.4bn/yr compliance cost



(1) Recalibrate Physical EAL (Return Period Compression)

DATA

MYS 100-yr flood: 216 mm | EAL gap +5.9%
PHL 100-yr storm: 521 mm | EAL gap +1.3%
PELT regime break: 2007 (both markets, synchronized)

ACTIONS

- MYS: +6–8% flood loading on XL layers (EAL gap +5.9%; +1–2pp for n=34 CI margin)
- PHL: +1–2% floor loading ONLY (CI [326–985mm] is 3× wide — no point-estimate)
- Replace burning cost with GEV EAL (treaties > USD 50M)

CHIRPS v2.0 · GEV MLE · ruptures PELT · EM-DAT 1991–2024

(2) Apply Country-Specific Transition Risk Surcharges

DATA

MYS: USD 22.4bn/yr
LULUCF emitter: +63 MtCO₂e (palm oil / deforestation)
PHL: USD 14.8bn/yr
LULUCF net sink: -27 MtCO₂e (reforestation credit)

ACTIONS

- MYS: 3–5% surcharge (BNM CCPT C3/C4 cedants)
Add Climate Warranty Clause at renewal
- PHL: 1–2% surcharge (LULUCF sink → Art. 6.2 ITMO credit partially offsets)
- Re-evaluate at every NGFS Phase update

NGFS GCAM 6.0 · BNM CCPT 2021 · BSP Circ.1085 · UNFCCC Art.6

(3) Pass-Through Governance & ENSO Forward Protocol

DATA

PT rate 1%→5% = 5× treaty exposure variance:
SEA: USD 0.37–1.86bn/yr
HRe (10% alloc): USD 3.0M–14.9M/yr
ENSO: ONI r = -0.016, p = 0.93 — absent at annual scale

ACTIONS

- Apply 5% PT to C3/C4 if carbon > USD 111/t
Review annually at each NGFS Phase release
- ENSO is a null finding, not a pricing driver
Use NOAA La Niña 12-mo outlook as TCFD facultative audit trigger — NOT a pricing variable

NGFS GCAM 6.0 · NOAA ONI 1950–2026 · BNM CCPT tiers